ELECTRONIC SPEED CONTROLLER

ARK 4IN1 ESC

NDAA-compliant 4-in-1 ESC running ARK32 firmware

The USA-built ARK 4IN1 ESC is a high-performance 4-channel electronic speed controller for brushless motors. Four independent STM32F051 MCUs drive a 3-phase power stage delivering 50 A continuous / 75 A burst per motor under ARK32 firmware (supports AM32).

An onboard 0.2 mΩ shunt and current-sense amplifier provide 100 A/V sensing. Configure KV and pole count for each motor before first use.

KEY FEATURES

- 50 A continuous / 75 A burst per motor
- 3–8S LiPo (6 V min, 65 V abs max)
- 4× independent STM32F051 channels
- Onboard current sensor (100 A/V)
- ARK32 firmware (supports AM32)
- DShot 300/600, bidirectional DShot, KISS telemetry, PWM
- 30.5 mm mounting pattern

SPECIFICATIONS

ELECTRICAL	
Input voltage	6–65 V (3–8S)
Continuous current	50 A per motor
Burst current	75 A per motor
Current sense	100 A/V
Firmware	ARK32 (supports AM32)
Operating temperature	−25 °C to +85 °C

MECHANICAL	
Dimensions	43.0 × 40.5 × 7.6 mm
Weight	14.5 g
Mounting	30.5 mm pattern



AT A GLANCE	
Voltage	3–8S (6–65 V)
Current / motor	50 A cont / 75 A burst
Firmware	ARK32 (AM32)
MCU	4× STM32F051
Sense	100 A/V
Dimensions	43 × 40.5 × 7.6 mm
Weight	14.5 g
Hardware rev	Rev 4.2

COMPLIANCE & SOURCING

- Designed and manufactured in the USA
- NDAA compliant
- Blue UAS Framework listed

ORDERING & RESOURCES

STORE arkelectron.com/product/ark-4in1-esc/

DOCS docs.arkelectron.com/products/electronic-speed-controller/ark-4in1-esc

CAD github.com/ARK-Electronics/ark-hardware

FLIGHT CONTROLLER CONNECTOR

PIN	SIGNAL	NOTES
1	VBAT	Battery voltage
2	CURR	Current sense output
3	TELEM	Serial telemetry
4	MOTOR 1	DShot / PWM
5	MOTOR 2	DShot / PWM
6	MOTOR 3	DShot / PWM
7	MOTOR 4	DShot / PWM
8	GND	

Connector: 8-pin JST-SH. Matches standard FC ESC pinout.

DEBUG CONNECTOR

PIN	SIGNAL	NOTES
1	3.3 V	
2–9	SWDIO / SWDCLK 1–4	Per-motor SWD
10	GND	

Connector: 10-pin JST-SH.

NOTES

- Before first use, configure KV and pole count for each motor.
- DShot is the recommended input protocol. PWM requires calibration.
- Motor and battery connections are solder pads.

VARIANT

PRODUCT	KEY DIFFERENCE
ARK 4IN1 ESC CONS	Connectorized (MR30 motors, battery cables); 30 A cont / 40 A burst limited by connectors; larger size / weight

ARK-DS-ARK-4IN1-ESC | Version 1.0, 2026-10-29 | Compiled from the ARK 4IN1 ESC Production schematic Rev 4.2, the Production bill of materials, manufacturing documentation, and ARK Electronics published documentation. Current ratings are board thermal limits. Temperature ratings per component data sheets; the JST-SH connectors set –25 °C. Specifications are subject to change without notice. © 2026 ARK Electronics LLC.